Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 October 2026
Team ID	EN25CS3010534
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Access the four project datasets (athletes.csv, medal_results.csv, games_summary.csv and alltime_country_rankings.csv) and integrate them using noc_code and year as relationship keys.	High
2	Data Processing	Check data types, nulls and duplicates in each file, confirm that every noc_code matches across the files, set geographic roles for the country field, and create a Year field for trend analysis.	High
3	Business Analysis	Calculate total, Gold, Silver and Bronze medals by country; average medals per athlete by sport category; total medals per year; average medal efficiency percentage by region; sports investment against medal performance; medal totals for host nations; and KPIs for athlete participation, sports categories entered, nations represented and regional efficiency.	High
4	Business Insights & Analytical Outputs	Provide two dashboards (Global Sports Performance Analytics and Olympic Performance and Regional Efficiency) with KPI cards, a world map, a line chart, bar charts, a pie chart, a bubble chart and a donut chart, plus a Tableau story that explains the main findings.	High
5	User Interaction & Analysis	Let users filter by Country, Year, Sport Category, Region and Host Country, switch a chart between Total, Gold, Silver and Bronze with the KPI Selector parameter, and use tooltips to compare values.	High
6	Data Update & Reporting	Refresh the CSV data sources in Tableau, republish the workbook to Tableau Public, and show the updated dashboards and story in the Flask website. Keep screenshots, documentation and a demo video in the project folder.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Charts have clear titles, data labels and simple wording. Filters are placed on the dashboard, so a viewer can explore the data without training. The website has one page for each dashboard and the story.
NFR-2	**Security**	Only the project datasets are used and published, and they hold no real personal or confidential records. There are no logins, and no passwords or keys are stored in the GitHub repository.
NFR-3	**Reliability**	The same CSV files always give the same results. The noc_code keys match across all four files, and the workbook is saved as a packaged workbook (.twbx) so the data travels with it.
NFR-4	**Performance**	The four files are small (8,000, 910, 18 and 60 rows), so Tableau loads them quickly and the dashboards respond to filters without delay.
NFR-5	**Availability**	Published dashboards are available through the Tableau Public link at any time. A demo video and screenshots are kept as a backup in case the live link or the local Flask site is not available.
NFR-6	**Scalability**	New Games, sports or countries can be added by updating the CSV files. The CSV data can later be moved to a database such as MySQL or Snowflake without changing the dashboard design.